

ANALYSIS OF QR CODE IMPLEMENTATION IN LIBRARY INFORMATION SYSTEM

Written to Fulfill The 2nd Task of Information Systems Architecture and Technology



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PREFACE

In the name of Allah SWT. Praise and gratitude for His presence, mercy, and guidance so that the author can complete this writing task well and smoothly. This paper has been compiled by the author to the best of his ability and with the assistance of several parties. Therefore, the author would like to express his gratitude to those who contributed to the completion of this writing assignment. The author realizes that there are still many shortcomings and errors in this writing assignment. Therefore, we are open to any criticism and suggestions from readers so that the author can improve the writing.

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CHAPTER I INTRODUCTION

I.1 Background

One of the important facilities in supporting educational and learning activities is the library. The library is a space that provides a variety of collections of books and reference materials that can be used by students, educators, and the general public to increase their knowledge and insight. However, library management that still relies on manual systems often creates operational problems.

Manual systems that are still commonly used in libraries make it difficult for library staff to document all activities such as member registration, visitor attendance records, and book borrowing and returns in a traditional manner. This manual recording process is quite time-consuming and is still considered highly prone to human error. In addition, records kept in notebooks are also very susceptible to damage, such as tearing, loss, incorrect writing, and illegible writing. This can cause data to become invisible and lost.

Other potential problems include difficulty in locating borrowed books, errors in calculating late fees, and the time required to compile reports due to having to manually collect information from logbooks. This situation results in suboptimal library service quality and hinders the productivity of library staff.

With the advancement of information technology today, a library information system is needed to solve these problems. One such system is the QR Code (Quick Response Code). A QR Code is a type of two-dimensional barcode that can store more data and can be read quickly using a smartphone camera or webcam. The application of QR Codes in library systems can speed up the process of identifying members and books, so that borrowing and returning transactions can be carried out more quickly and accurately.

Referring to the existing background, an analysis was conducted on the application of QR codes in library information systems to understand how this technology can improve the efficiency and quality of library services.

I.2 Writing Objectives

The purpose of this study is to identify the technology used in QR Code-based library information systems, analyze the architecture of library information systems that use QR Codes, and evaluate the advantages and challenges of implementing QR Codes in library systems.

I.3 Problem Domain

The object of analysis in this paper is a web-based library information system that uses QR codes, with a literature study from journals on the implementation of QR code-based library systems. This study examines how QR codes are implemented in library information systems, identifies the technologies used in QR code-based library information systems, analyzes the architecture of library information systems that use QR codes, and evaluates the advantages and challenges of implementing QR codes in library systems.

I.4 Writing Methodology

The type of research methodology used in this study is using a literature study approach. The sources of this paper are scientific journals, articles, and publications discussing the implementation of QR Code-based library information systems.

I.5 Writing Framework

Chapter I : Introduction

Contains the background of the problem, purpose of writing, scope of discussion, writing methodology, and writing framework.

Chapter II : Basic Theory

Explaining the basic theories that form the basis of analysis, such as the QR Code library information system, including its definition, how it works, and its advantages.

Chapter III : Problem Analysis

Problems with manual library systems Implementation of QR codes as a solution technologies used include recommendations, explanations of QR code-based library system architecture and system features and functions, advantages and challenges of implementation, and evaluation of success.

Chapter IV : Conclusion and Suggestion

Contains conclusions from the analysis results and suggestions for further development.

CHAPTER II BASIC THEORY

Before analyzing the application of QR codes in library systems, it is essential to first understand the basic concepts underlying this system. This chapter will discuss theories related to information systems, library information systems, QR Code technology, web technology used, and system architecture. Understanding these concepts is an important part of analyzing the application of QR Codes in library systems, which will be explained in Chapter 3.

II.1 Information System

An information system is essentially an interconnected ecosystem that links a series of work processes with technology and human activities, with the aim of transforming raw data into insights that support managerial decisions. As a dynamic entity, this system functions with a distinctive feature whereby various elements within it are interconnected in a specific area but still interact with the external environment through a mobile interface, to process each input into an output that is in line with the organization's objectives. The continuity of this system's function depends entirely on the harmonious collaboration of five important elements, namely hardware and software as the technological foundation, databases as the center of information, procedures as operational guidelines, and humans as the main managers who run this system. The information system can be implemented in the many areas such as library information system. The information system consist many technology like QR code.

II.2 QR Code

A Quick Response Code (QR Code) is a two-dimensional pattern that functions as a large-capacity digital information storage device, including links (URL), text, and contact information, and is designed to be quickly read using scanning devices such as mobile phones. Unlike one-dimensional barcodes, QR Codes offer extremely fast scanning speeds and greater resistance to damage. In terms of utility, this technology has become an effective identification and tracking method in various fields, allowing users to access detailed information with just one scan, which in turn speeds up transactions, document verification, and various other operational activities.

II.2.1 Main Component

In performing its role, QR Code has several components that support the scanner's work, namely:

A. Positioning Detection Markers.



Image II.3.1 Positioning Detection Markers (Ref: <https://glints.com/id>)

The first component is called Positioning Detection Markers, which are three square-shaped elements within the QR Code. They are always located in the corners of the QR Code. This component ensures that the scanner can quickly read the code and determine its orientation or position.

B. Alignment Marking



Image II.3.1 Alignment Marking (Ref: <https://glints.com/id>)

The difference between the first and second components is that the second one is smaller than the position detection markers but has the same square shape. This component serves to maintain the surface of the QR Code even when printed on a curved surface. The more data stored in a QR Code, the larger its size. Therefore, the number of components can also be greater.

C. Timing Pattern



Image II.3.1 Timing Pattern (Ref: <https://glints.com/id>)

The timing pattern on a QR Code is the part that looks like a series of squares. This pattern functions as a data grid configuration. With this component, QR Code scanners are able to determine the size of the data matrix being loaded.

D. Version Information



Image II.3.1 Version Information (Ref: <https://glints.com/id>)

The fourth component provides information about the QR Code version. To date, there are 40 different types of QR Codes. This mark allows the scanner to identify which version of the QR Code is being scanned. Generally, versions 1 to 7 are the most commonly used.

E. Format Information.



Image II.3.1 Format Information. (Ref: <https://glints.com/id>)

Format Information is a component that explains error tolerance and data mask patterns. With format information, scanners can more easily scan QR Codes to display the data they contain to users.

F. Data and Error Correction Keys



Image II.3.1 Data and Error Correction Keys (Ref: <https://glints.com/id>)

This area of the QR Code is important to have in the code structure because it is where all the data is stored. In addition, this area or section also includes an error correction block that keeps the data scannable even if the code is damaged by up to 30%.

G. Quiet Zone



Image II.3.1 Quiet Zone (Ref: <https://glints.com/id>)

The Quiet Zone is the empty space located on the outer edge of a QR Code. Similar to white space in design, the Quiet Zone is an important element in QR Codes. This is because the Quiet Zone in a QR Code serves to emphasize its structure and make it easier to scan. The Quiet Zone must be present to separate the QR Code from its surroundings so that scanners can recognize it without difficulty. Although empty, this area is a vital part of a QR Code.

II.2.2 Scanning Process

QR codes are square-shaped and their patterns represent binary codes that are read to reveal code data. They consist of black dots, black blocks, and spaces. QR readers on mobile phones can identify standard QR codes based on the three large squares outside the QR code to verify that all patterns are QR codes. QR readers analyze QR codes by breaking down all patterns into squares. It identifies the black and white patterns in the squares and groups the squares into larger patterns to scan the data, information, and meaning. Black patterns represent “1” and white patterns

represent binary “0”. The scanning process will be explained further below. Here are the scanning processes:

A. Open the Scanner Application.

You can open the scanner application using the QR Reader on your mobile phone or a mobile phone camera equipped with a QR Reader feature.

B. Point the camera at the QR Code.

Then point the app or camera at the QR Code. The scanner will detect three large boxes in the corners to determine the position and orientation.

C. Data Reading.

After the QR Code detects three large boxes in the corners, the scanner will read the pattern of black and white dots arranged in the boxes, then process the information stored in them.

D. Information Display

After all steps are completed, the information will appear on the mobile phone screen. It can be in the form of a website link, text, contact details, or other information.

II.2.3 Storage Capacity

QR codes can store a large amount of data to display various types of information. QR codes can store 2089 digits or around 4289 characters, including punctuation marks and special characters. This makes QR codes much more powerful than regular barcodes.

II.2.4 Error Collection Feature

QR Code error correction can reach 30%, QR codes can still be scanned and the information directed from the related code is also displayed accurately. This is thanks to an error correction system that has several levels (L, M, Q, and H). So, simply put, QR Codes work by storing information in visual patterns that can be read and interpreted by a cell phone camera into usable data.

II.3 Web Technology

The development of web-based library information systems is highly dependent on the use of a combination of server-side and client-side technologies to create interactive and dynamic applications. As a foundation, PHP (Hypertext Preprocessor) is used as the main programming language for the server side, which is responsible for processing business logic and interacting

with databases, while JavaScript is used on the client side to enhance the aspects of interactivity in the user interface, for example in form validation or page element manipulation. To accelerate development progress, frameworks such as CodeIgniter are used to implement MVC (Model-View-Controller) architecture. This form effectively separates the Model (data logic), View (display), and Controller (processing logic), so that the code structure becomes more organized and easier to maintain. All key data, from member information to book collections, is managed by MySQL, a Relational Database Management System (RDBMS) that handles large amounts of data well and supports access by multiple users. In addition, to support automation, the system includes integration with a QR Code Library that enables the automatic creation and scanning of unique codes, such as those affixed to each copy of a book, to facilitate the circulation and inventory processes.

II.4 System Architecture

The system architecture design was carried out by applying the MVC (Model-View-Controller) design pattern. This pattern aims to divide application components into three main elements that are functionally separate: Model (which manages data and business logic), View (which manages the display for users), and Controller (which acts as a connector and manages interactions between Model and View). This separation is designed to facilitate the development process, maintenance, and the system's ability to evolve in the future. To visualize the structure and workflow of the system prior to the implementation phase, UML (Unified Modeling Language) is used as a standard modeling tool that clearly shows the relationships between components.

CHAPTER III PROBLEM ANALYSIS

III.1 Library Information System

A library information system is an application of information technology designed to manage library circulation and administration, replacing manual processes with computer-based ones. The purpose of implementing this system is to meet the needs of processing collection and member data, as well as to support service operations so that information can be delivered promptly to users.

The main objective of this system is to improve the quality of service to members, where the system needs to be managed by professionals who understand literature management to facilitate reference searches. With an organized system, ease and speed of access are prioritized so that the best service can be provided and the information needs of library visitors can be met.

III.2 Analysis of Manual Library System Issues

Based on the analysis of the three journals, the same problem was found in the manual library system. In the first case study, library data management still uses manual methods in daily activities, where the process of recording inventory when new books arrive takes a long time because it must be written repeatedly even though the books have the same identity. The work is time-consuming and overly complicated, as is the case with services such as recording visitor lists and the process of lending books. Searching for book and visitor data requires reopening the master book page by page to find the desired data, which takes a considerable amount of time.

The same problem also occurred in the second case study. In the library service process, registered members must fill out an attendance book and leave their membership cards with the staff. This process is very time-consuming, and members often do not write in the attendance book, making it difficult for staff to accurately determine visitor numbers. When recording the borrowing and returning of books, errors often occur, such as members being charged fines that do not match the amount of the fine incurred, or clerks making mistakes in recording transactions.

In the third case study, the library still uses manual methods to manage book borrowing data and weekly or monthly reports. Data storage media for books still uses ledgers, which are often

damaged, torn, or lost, resulting in the loss of the data contained therein. Manual recording errors often occur due to human error. In addition, the length of time it takes to record new books because manual methods are still used is an obstacle to library operations.

From these three case studies, we can conclude that the main problems with manual library systems include slow and time-consuming processes, high rates of human error in data recording, damage to or loss of physical data in the master book, difficulty in tracking the location of books, lengthy inventory processes, and time-consuming report generation that is highly prone to error. These problems indicate that the manual system is inefficient and needs to be replaced with a technology-based system to improve the quality of library services.

III.3 Solution analysis: QR Code implementation

QR codes can be implemented as a technological solution to overcome manual system problems in libraries. In this implementation, QR codes are used for two main functions: member identification through membership cards and book collection identification through book labels. Each library member receives a card with a unique QR code containing their member ID. The QR Code on the membership card can be used for checking in and out of the library and for identification during book borrowing transactions. In addition, each book is given a QR Code label containing the book ID or detailed book information. The QR Code label on the book is used to identify the book, to display detailed book information, and to speed up borrowing and return transactions.

The system in the first case study implements QR codes on book labels and membership cards using a scanner to scan the QR codes. The system in the second case study also implements QR codes for membership cards and books. In the application created, officers do not need to input much data because the data that has been input is replaced by scanning. When a member returns a book, the staff only needs to scan the membership card and the borrowed book, after which the data will appear automatically along with the fines. Meanwhile, in the third case study, the use of QR codes is applied in the book borrowing transaction process to support the speed of the system and the accuracy of the data in a book.

The QR Code-based library system workflow begins with the member registration process, where the staff inputs member data. The system then automatically generates a QR Code, and the membership card can be printed. For attendance, members simply scan the QR Code on their membership card, the system validates it, and attendance data is automatically stored. The borrowing process is carried out by the staff first scanning the QR Code on the membership card, then scanning the QR Code of the book to be borrowed. The system will validate it according to applicable borrowing rules, and the transaction will be stored in the database. For the return process, the staff scans the QR Code.

For returned books, the system will calculate the loan duration and any late fees, then update the book's status to "available again."

With the implementation of QR codes, a previously manual and extremely slow process becomes automated and incredibly fast. A single scan instantly retrieves data from the database, reducing service time from several minutes to just seconds. This significantly improves library operational efficiency and reduces queues for patrons seeking book purchases.

III.4 Analysis of Technology Used

Based on an analysis of the three journals, there are variations in the technologies used to build QR Code-based library information systems. In terms of backend programming languages, the first case study uses the Java Script programming language with the ReactJS library, a modern approach with real-time capabilities and a responsive user interface.

The second case study uses the native PHP programming language, which is simple and easy to learn, making it ideal for small to medium-sized libraries. Meanwhile, the third case study uses the PHP-based CodeIgniter framework, which offers an organized code structure using the MVC (Model View Controller) pattern, accelerating the development process and simplifying maintenance.

Of these three approaches, using a PHP framework like CodeIgniter or Laravel can be considered the best choice for implementing medium to large library systems. Frameworks offer a balance between ease of development and good code structure. The resulting code is more structured, easier

to maintain, and supports system scalability. While there's an initial learning curve, the long-term benefits in terms of development efficiency and code quality are significant. Native PHP is suitable for small libraries with simple feature requirements, while JavaScript with ReactJS is more suitable for highly interactive and modern user interfaces, despite its higher complexity.

For database technology, two approaches were used. The first case study used Cloud Firestore from Firebase, a cloud-based NoSQL database with real-time synchronization capabilities. Cloud Firestore offers advantages in deeper scaling and keeps data connected across client applications through real-time listeners. Meanwhile, the other two case studies used MySQL as the database management system. MySQL is a structured, mature, and widely used relational database.

For standalone library systems that don't require real-time multi-location access, MySQL can be considered the best choice. It's highly stable, doesn't rely on an internet connection, is free to use, and boasts extensive documentation and a community. Relational databases like MySQL are also better suited for library data structures that have clear relationships between entities such as members, books, and transactions. Cloud Firestore is more suitable if the library system needs to be accessed from multiple locations in real time or if there's a need for instant data synchronization between devices.

For QR Code technology, all systems use similar libraries to generate and scan QR Codes. QR Code generation is done using libraries such as `phpqrcode` for PHP or `qrcode.js` for JavaScript. These libraries function to convert data (such as member IDs or book IDs) into QR Code images through an encoding process. To read QR Codes, scanner libraries such as `Html5-QRCode` or `ZXing` are used, which can access the device's camera (webcam or smartphone) to read the QR Code pattern and decode it back into data. The way QR Code technology works is quite simple but effective: data is encoded into a QR Code image when creating a card or label. Then, when needed, the camera reads the QR Code, the system decodes the QR pattern into ID data, and uses that ID to query the database to display complete information.

III.5 System Architecture Analysis

The QR code-based library information system analyzed from the three journals uses a three-layer architecture, a best practice in web application development. This architecture divides the system into three main layers, each with separate but integrated responsibilities and functions.

The first layer is the Presentation Layer, the display part that interacts directly with the user. At this level, the system is displayed through a web browser using HTML, CSS, and JavaScript, including a QR Scanner feature that utilizes a laptop or smartphone camera. All user activities, such as filling out forms, pressing buttons, or scanning QR codes, are processed in this layer and then forwarded to the next layer via HTTP requests.

The second layer is the Business Logic Layer, which handles all core system processes. In some implementations, this layer can be created with JavaScript or PHP, with or without a framework. The primary tasks of this layer include managing business logic, such as the login process and user access rights, generating QR codes for membership cards and book labels, checking and validating QR codes when scanned, and implementing loan rules, including calculating fines. This layer also provides an API that bridges communication between the user interface and the database. All data processing, system decisions, and work rules are executed in this layer.

The third layer is the Data Layer, which is responsible for permanently storing all data. In practice, this layer can use either MySQL or Cloud Firestore. The database stores various types of data, such as member information, book collections, loan and return records, and visitor attendance logs. This layer interacts with the Business Logic Layer through SQL commands (if using MySQL) or API calls (if using Firestore).

One of the biggest advantages of a three-layer architecture is the clear separation of functions between each layer. The presentation layer handles only the display and user interaction, the business logic layer processes data and enforces system rules, while the data layer focuses on data storage and retrieval. This division of roles simplifies maintenance because changes to one layer won't affect the others as long as the communication path remains the same. For example, changing the interface design won't affect the system logic or database structure. Furthermore, a three-layer architecture supports scalability because each layer can be increased in capacity independently.

Security-wise, this model is more secure because the logic and data are separated from the display, preventing users from directly accessing both.

III.6 System Feature and Function Analysis

A review of the three journals reveals that QR Code-based library information systems generally share a similar set of core features. One such feature is member data management, where staff can add new members and the system automatically generates a QR code for use on membership cards. These cards can be printed if needed. The advantage is that member identification is digital, so when changes occur, the card doesn't need to be reprinted; a new QR code can simply be generated through the system.

The visitor attendance feature allows attendance to be recorded by scanning the QR code on the membership card. In one study, the system provided a dedicated page for this attendance. All attendance data is recorded directly and stored accurately, allowing staff to view reports or graphs of visitor numbers without the need for manual recaps from the guest book. This method minimizes issues encountered with conventional systems, such as visitors frequently forgetting to fill out attendance lists.

The QR Code-based book catalog feature makes it easier for visitors to quickly obtain book information. Simply by scanning the QR Code on the book label, the system displays complete information such as title, author, publisher, year of publication, category, and shelf location. This feature eliminates the need for frequent inquiries from staff when searching for information, enhancing the user experience and reducing staff workload.

The borrowing feature has become much more efficient with the use of QR codes. The clerk simply scans the QR code on the member's card to identify the borrower, then scans the QR code of the book they wish to borrow. The system will automatically perform checks, such as ensuring the member still has a loan quota, no outstanding fines, and verifying the book's availability. The entire process is very fast, taking only about 10–15 seconds, compared to manual methods that typically take several minutes.

The book return feature has become much more convenient thanks to the use of QR codes. Staff simply scan the QR code on the returned book, and the system will calculate the loan period and determine if there are any overdue days. If late, the system automatically calculates a fine based on the number of days overdue multiplied by the predetermined rate. This eliminates the potential for calculation errors common with manual methods. Once the process is complete, the book's status is immediately updated to available, allowing other members to borrow it.

The reporting feature allows the system to automatically generate various types of reports. The system can generate reports on visitor data, membership, book collections, inventory, copies, and even circulation reports. All these reports can be exported to PDF or Excel, making them easier to use for archiving and presentation purposes. This feature is very helpful for staff, as processes previously performed manually through ledgers can now be completed much more quickly and efficiently.

Analysis of the book borrowing process shows that the use of the QR Code system provides a significant increase in efficiency. With the manual system, staff needed approximately 3–5 minutes to find the membership card in the filing cabinet, write down the loan details in a notebook or on the book slip, and then re-enter the details on the membership card. After switching to QR Codes, all these steps can be completed in just 10–15 seconds, simply by scanning two QR Codes and the system immediately storing all the necessary information. This increased efficiency by approximately 95% and directly impacted the reduction of queues and increased user experience.

CHAPTER IV CONCLUSION AND SUGGESTION

IV.1 Conclusion

Based on the analysis conducted on three case studies of the implementation of QR Code-based library information systems, it can be concluded that manual library systems have various fundamental problems that hinder the operational efficiency of libraries. These problems include slow and time-consuming processes, high levels of human error in data recording, damage or loss of physical data in the master book, difficulty in tracking the location of books, lengthy inventory processes, and time-consuming and error-prone report generation. These problems indicate that manual systems are inefficient and need to be replaced with technology-based systems.

The implementation of QR codes has proven to be an effective solution to overcome the problems of manual library systems. QR codes are used in two main functions, namely as member identification through membership cards and as book collection identification through book labels. The QR code-based system is capable of automating manual processes such as visitor attendance, book borrowing, and book returns, thereby significantly increasing the speed and accuracy of services.

IV.2 Suggestion

Further research is needed in five important areas to improve the QR Code-based library system. First, a comparison study should be conducted to evaluate how effective QR Code is compared to other identification technologies like RFID and NFC. This will provide insights into the strengths, weaknesses, and best conditions for each technology. Second, long-term research on the impact of this system on library visitors is important, especially to measure increases in visit frequency and book borrowing numbers. Third, a more detailed cost-benefit analysis for libraries of different sizes should be carried out to help them make better-informed investment decisions. Fourth, research on user experience and the design of the most user-friendly interface should be done to identify the most commonly used features and develop an improved system. Fifth, security and privacy aspects in the QR Code-based library system need to be considered, including protection of members' personal data and prevention of misuse of membership cards to build user trust in the system.

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